

BORCEA'S 2-VARIANCE CONJECTURE

TENG ZHANG

ABSTRACT. We prove that every zero of a complex polynomial of degree at least two lies at distance at most the standard deviation of its zeros from a critical point. Zeros are counted with multiplicity, and the constant is optimal. This establishes Borcea's variance conjecture for every exponent $p \geq 2$.

1. INTRODUCTION

The Gauss–Lucas theorem places the critical points of a complex polynomial in the convex hull of its zeros. It leaves open a different question: how close must a critical point lie to a prescribed zero? Sendov's conjecture, proposed in 1958 and publicized by Hayman [Hay67, Problem 4.5], asserts that distance one suffices when all the zeros lie in the closed unit disk. The classical setting and its many variants are treated by Marden [Mar66] and Rahman and Schmeisser [RS02]. Among the principal advances, Meir and Sharma [MS69] settled degree five, Borcea [Bor96] treated polynomials with at most seven distinct zeros, Brown and Xiang [BX99] settled degrees at most eight, Dégot [Deg14] treated sufficiently large degrees with the prescribed zero fixed in the disk, and Tao [Tao22] obtained a degree threshold independent of that zero. More recently, Mazur [Maz26] gave a computer-assisted proof, subsequently reorganized and explained by Tao [Tao26].

Borcea proposed measuring the spread of the zeros by their centered moments instead of the radius of a containing disk. If P has degree $n \geq 2$ and zeros $z_1, \dots, z_n$, counted with multiplicity, define

$$\begin{aligned}\sigma_p(P) &= \inf_{c \in \mathbb{C}} \left(\frac{1}{n} \sum_{j=1}^n |z_j - c|^p \right)^{1/p} \quad (1 \leq p < \infty), \\ \sigma_\infty(P) &= \inf_{c \in \mathbb{C}} \max_{1 \leq j \leq n} |z_j - c|.\end{aligned}\tag{1.1}$$

Write $Z(P)$ for the set of zeros of P and put $d(P) = \max_{a \in Z(P)} \min_{\zeta \in Z(P')} |a - \zeta|$. Khavinson, Pereira, Putinar, Saff, and Shimorin [KPPSS11, Definition 2.1 and Conjecture 1] recorded Borcea's conjecture in the following form.

Conjecture 1.1. *For every complex polynomial P of degree at least two and every $p \in [1, \infty]$, one has $d(P) \leq \sigma_p(P)$.*

The case $p = \infty$ is equivalent to Sendov's conjecture by translation and dilation. For $p = 2$, the minimizing center is the centroid $c_P = n^{-1} \sum_j z_j$, and

$$\sigma_2(P)^2 = \frac{1}{n} \sum_{j=1}^n |z_j - c_P|^2.\tag{1.2}$$

Thus the quadratic case asks for a bound in terms of the standard deviation of the zeros. Borcea [Bor07, Theorem 4] proved a variance bound for the distance from an equilibrium point of a logarithmic potential to the charges. For equal charges, this controls the distance from a critical point to the zeros, whereas Conjecture 1.1 prescribes a zero and seeks a nearby critical

2020 *Mathematics Subject Classification.* Primary 30C15; Secondary 26D15, 65G30.

Key words and phrases. Zeros of polynomials, critical points, Borcea's variance conjecture, Schoenberg's inequality, computer-assisted proof.

point. Khavinson, Pereira, Putinar, Saff, and Shimorin [KPPSS11, Theorems 3.2 and 6.1, and §4] proved the quadratic conjecture for polynomials with at most three distinct zeros. Their account also presents Borcea’s proofs of the $p = 1$ conjecture for cubic polynomials and a stronger quadratic bound for polynomials with real zeros. They also developed matrix formulations of the general problem and described its origins in Borcea’s unpublished work.

Our main result settles the quadratic conjecture.

Theorem 1.2. *Let P be a complex polynomial of degree $n \geq 2$. For every $a \in Z(P)$ there exists $\zeta \in Z(P')$ such that*

$$|a - \zeta| \leq \sigma_2(P). \quad (1.3)$$

The constant 1 in (1.3) is optimal.

Corollary 1.3. *Conjecture 1.1 holds for every $p \in [2, \infty]$.*

Proof. For each fixed center, the normalized power means are nondecreasing in p . Taking infima over the centers in (1.1) gives $\sigma_2(P) \leq \sigma_p(P)$ for $p \geq 2$. The assertion follows from Theorem 1.2. $\square$

The proof of Theorem 1.2 is computer-assisted. Degrees $n \geq 100001$ are treated analytically. For $4 \leq n \leq 100000$, two certificates partition a bounded three-dimensional parameter region into 33671 boxes. Exact rational arithmetic and outward-rounded integer bounds exclude each box. The certificates, checkers, and complete logs are available in the supplementary repository [Zha26S]. The finite computation checks inequalities on whole boxes; it does not use numerical approximations to polynomial zeros.

We describe the relation with the preceding work more precisely. Schoenberg [Sch86] conjectured a sharp second-moment inequality for the critical points; Pereira [Per03] and Malamud [Mal05, Theorem 4.15] proved it independently. This inequality supplies the critical-point variance bound used here. The reciprocal coordinates and antiderivative identities follow Tao [Tao26, Lemma 6 and the proof of Proposition 11]. We also use the centered interpolation of Zhang [Zha26, Lemma 5.1]. Zhang’s paper proves the quadratic Tang–Zhang inequality, proposed in the work of Tang and Zhang [TZ25, Conjecture 1.10]: for zeros in the closed unit disk and a prescribed zero a , the sum of the reciprocal squared critical-point distances is at least $n - 1$. That disk hypothesis is stronger than a bound on the variance and does not follow from the normalization (1.2). For example, a zero at $\sqrt{n-1}$ and $n-1$ zeros at $-1/\sqrt{n-1}$ have variance one, while the radius of their smallest containing disk exceeds one when $n > 2$. Conversely, our argument assumes that every critical-point distance exceeds the standard deviation; it does not establish the corresponding reciprocal-sum inequality under a variance hypothesis.

The replacement for the disk condition is a weighted covariance estimate in reciprocal coordinates, proved in Lemma 3.1. It controls both the endpoint of the antiderivative integral and the variance of the reciprocal coordinates. The integral estimates then exclude all sufficiently large degrees. For the remaining degrees, product bounds give further necessary conditions, while a centered coefficient expansion treats configurations close to equality. In particular, retaining the real part of the quadratic coefficient excludes an entire neighborhood of the limiting configuration; the finite covering therefore has no unresolved limiting region.

Section 2 gives the normalization and the notation used throughout the proof. Section 3 establishes the covariance inequalities and polynomial identities. Sections 4 and 5 develop the integral estimates and treat large degrees. Sections 6 and 7 provide the additional estimates for small degrees. Section 8 completes the proof. Appendices A and B specify the bounds used on each parameter box and the exact arithmetic used to evaluate them.

2. NORMALIZATION AND NOTATION

All zeros and critical points are counted with multiplicity. We first state the two results used in the reduction. The centered form of Schoenberg’s inequality is as follows.

Lemma 2.1 ([Mal05, Theorem 4.15]). *Let Q have degree $d \geq 2$, zeros $\lambda_1, \dots, \lambda_d$, and critical points $\eta_1, \dots, \eta_{d-1}$. If $\sum_{j=1}^d \lambda_j = 0$, then*

$$d \sum_{j=1}^{d-1} |\eta_j|^2 \leq (d-2) \sum_{j=1}^d |\lambda_j|^2. \quad (2.1)$$

The following result includes degrees two and three.

Lemma 2.2 ([KPPSS11, Theorem 3.2]). *If a complex polynomial P of degree at least two has at most three distinct zeros, then $d(P) \leq \sigma_2(P)$.*

Suppose that (1.3) fails at a zero of a polynomial of degree $n = m + 1 \geq 4$. The variance is positive and the prescribed zero is simple. Translate the centroid to zero, divide the zeros by $\sigma_2(P)$, rotate the prescribed zero to the nonnegative real axis, and make the polynomial monic. The resulting polynomial satisfies

$$\begin{aligned} P(z) &= (z - a) \prod_{j=1}^m (z - z_j), \quad a \geq 0, \\ a + \sum_{j=1}^m z_j &= 0, \quad a^2 + \sum_{j=1}^m |z_j|^2 = m + 1. \end{aligned} \quad (2.2)$$

Let $\zeta_1, \dots, \zeta_m$ be its critical points. Coefficient comparison and (2.1) give

$$\frac{1}{m} \sum_{j=1}^m \zeta_j = 0, \quad V_* := \frac{1}{m} \sum_{j=1}^m |\zeta_j|^2 \leq V_m := 1 - \frac{1}{m}. \quad (2.3)$$

A counterexample has $|a - \zeta_j| > 1$ for every j , so (2.3) excludes $a = 0$. Cauchy–Schwarz in (2.2) gives $a^2 \leq m$.

In the rest of the paper, a *normalized counterexample* means a polynomial satisfying (2.2) with $a > 0$ and $|a - \zeta_j| > 1$ for every j . Unless stated otherwise, the following notation refers to such a counterexample.

- The symbol $\mathbb{E}$ denotes the average over $j = 1, \dots, m$: thus $\mathbb{E}f(q) = m^{-1} \sum_{j=1}^m f(q_j)$. For a real vector Y , set $\text{Var}(Y) = \mathbb{E}(Y - \mathbb{E}Y)^2$.
- The reciprocal coordinates, their mean, and their second moments are

$$\begin{aligned} q_j &= (a - \zeta_j)^{-1}, \quad \mu = \mathbb{E}q = x + iy, \quad r = |\mu|, \\ w_j &= q_j - \mu, \quad s = \mathbb{E}|q|^2, \quad v = \mathbb{E}|w|^2 = s - r^2, \quad Q_0 = \prod_{j=1}^m |q_j|^2. \end{aligned} \quad (2.4)$$

Here $x, y \in \mathbb{R}$, $0 < |q_j| < 1$, $0 < s < 1$, $0 \leq r < 1$, and $v \geq 0$. We also write $X_j = |q_j|^2$ and $\Delta = 1 - s$.

- The symbol V denotes a specified upper bound for V_* with $0 \leq V \leq 1$. Its default value is V_m from (2.3); individual arguments may use $V = 1$ or a sharper bound.
- For $t \in [0, 1]$, put

$$\rho = 2ax - a^2s, \quad \beta(t) = \mathbb{E}|1 - atq|^2 = 1 - \rho t - a^2s t(1 - t). \quad (2.5)$$

We shall also use

$$F(t) = \prod_{j=1}^m (1 - atq_j), \quad J = \prod_{j=1}^m z_j q_j, \quad h = a\mu, \quad D(t) = 1 - ht. \quad (2.6)$$

- For a vector $Y = (Y_1, \dots, Y_m)$, write $e_k(Y) = \sum_{1 \leq j_1 < \dots < j_k \leq m} Y_{j_1} \cdots Y_{j_k}$ and $e_0(Y) = 1$. Empty sums are zero and empty products are one. For $u \in \mathbb{R}$, write $u_+ = \max\{u, 0\}$.

The notation $a, z_j, \zeta_j, q_j, F, J$ follows Tao [Tao26]; the centered variables w_j and the reciprocal moments follow Zhang [Zha26, §2]. We use μ for the complex mean so that $m = n - 1$ remains available for the number of critical points.

Figure 1 shows how the reductions lead to the two degree ranges in the proof.

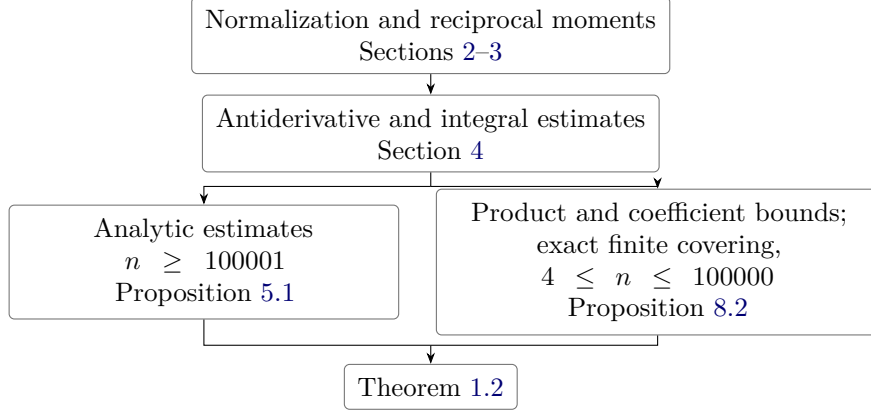


FIGURE 1. The two parts of the contradiction argument. Degrees two and three follow from Lemma 2.2.

3. RECIPROCAL MOMENTS AND POLYNOMIAL IDENTITIES

The identities in this section require only the normalization and the failure of the desired distance bound. In particular, no containing disk for the original zeros is assumed.

3.1. Covariance inequalities. The inequalities below use the positive weights $1 - |q_j|^2$. Their positivity follows from the failure of the desired distance bound, and is the point at which that hypothesis enters the covariance argument. Recall ρ and β from (2.5).

Lemma 3.1. *Every normalized counterexample satisfies*

$$\rho \geq r^2 + (1 - V)(1 - s), \quad |\mu - a|^2 \leq (a^2 + V - 1)(1 - s), \quad (3.1)$$

$$|1 - a\mu|^2 \leq Vv, \quad s(a^2 + V) \geq 1, \quad (3.2)$$

$$\beta(1) \leq (a^2 + V)v. \quad (3.3)$$

Proof. Since $q = (a - \bar{\zeta})|q|^2$ and $\mathbb{E}\zeta = 0$, $\mu - as = \mathbb{E}[\bar{\zeta}(1 - |q|^2)]$. The weights $1 - |q_j|^2$ are positive. Weighted Cauchy–Schwarz therefore gives

$$|\mu - as|^2 \leq \mathbb{E}[|\zeta|^2(1 - |q|^2)](1 - s) \leq (V - \beta(1))(1 - s),$$

where $\zeta q = aq - 1$ was used in the last step. Since $|\mu - as|^2 = r^2 - s\rho$ and $\beta(1) = 1 - \rho$, this proves the first inequality in (3.1). Expanding $|\mu - a|^2$ gives the equivalent second inequality.

Next, $a\mu - 1 = \mathbb{E}[\zeta(q - \mu)]$ proves the first inequality in (3.2). Applying Cauchy–Schwarz to q and $a - \zeta$ gives $1 \leq s\mathbb{E}|a - \zeta|^2 \leq s(a^2 + V)$, proving the second. Finally, $\beta(1) = |1 - a\mu|^2 + a^2v$ gives (3.3). $\square$

3.2. The antiderivative identities. Recall F, J, h, D from (2.6). The first identity below is Tao’s antiderivative identity [Tao26, Lemma 6, equation (9)]. The differentiated identity occurs in Tao’s proof of Proposition 11 [Tao26] and in Zhang [Zha26, Lemma 3.1]. We include the derivation because these identities continue to hold without the disk hypotheses of those papers. Define the remainder

$$\mathcal{R} = a^2 \int_0^1 t \sum_{j=1}^m q_j^2 \prod_{k \neq j} (1 - atq_k) dt. \quad (3.4)$$

Lemma 3.2. *For every normalized counterexample,*

$$(-1)^m J = (m+1) \int_0^1 F(t) dt, \quad (3.5)$$

$$1 = F(1) + mh \int_0^1 F(t) dt + \mathcal{R}. \quad (3.6)$$

One also has

$$1 = (-1)^m hJ + (1-h)^{m+1} - (m+1)h \int_0^1 (F(t) - D(t)^m) dt. \quad (3.7)$$

Proof. The factorization $P'(z) = (m+1) \prod_j (z - \zeta_j)$ gives $P'(a(1-t)) = (m+1)F(t)/\prod_j q_j$. Integrating from 0 to 1 and using $P(a) = 0$ and $P(0) = (-1)^{m+1}a \prod_j z_j$ proves (3.5).

Differentiating the product F gives $F'(t) + mhF(t) = -a^2 t \sum_j q_j^2 \prod_{k \neq j} (1 - atq_k)$. Integrating and using the definition (3.4) proves (3.6). The polynomial identity $(m+1)h \int_0^1 D(t)^m dt = 1 - (1-h)^{m+1}$, valid also when $h = 0$, together with (3.5), proves (3.7). $\square$

We shall use both a root product bound and a critical-point product bound.

Lemma 3.3. *Define*

$$\Phi_m(a) = a \left(1 + \frac{1-a^2}{m} \right)^{m/2}, \quad \psi(a) = a \exp \left(\frac{1-a^2}{2} \right).$$

Every normalized counterexample satisfies

$$|hJ| \leq \Phi_m(a) r \sqrt{Q_0}, \quad \Phi_m(a) \leq \min\{1, \psi(a)\}, \quad (3.8)$$

$$\left[\frac{m+1}{m} (1+a^2)s \right]^m \geq (m+1)^2. \quad (3.9)$$

For fixed a , $\Phi_m(a)$ is nondecreasing in m on its domain $m \geq a^2$. For fixed m , it increases up to $a = 1$ and decreases thereafter.

Proof. The arithmetic–geometric mean inequality in (2.2) gives $a \prod_j |z_j| \leq \Phi_m(a)$. Applying the same inequality to the $m+1$ nonnegative numbers $a^2, (m+1-a^2)/m, \dots, (m+1-a^2)/m$ proves $\Phi_m(a) \leq 1$. The inequality $\log(1+u) \leq u$ gives the bound by $\psi(a)$. Substituting these bounds into $|hJ| = ar \prod_j |z_j| \sqrt{Q_0}$ proves (3.8).

Set $u_j = a - z_j$. Then $\mathbb{E}|u|^2 = (m+1)(1+a^2)/m$ and $\prod_j |u_j|^2 = |P'(a)|^2 = (m+1)^2/Q_0$. The inequalities $Q_0 \leq s^m$ and $\prod_j |u_j|^2 \leq (\mathbb{E}|u|^2)^m$ prove (3.9).

For monotonicity in m , differentiate $m \log(1+b/m)$ with $b = 1-a^2$; the derivative is $\log(1+b/m) - (b/m)/(1+b/m) \geq 0$. Monotonicity in a follows by differentiating $\log \Phi_m(a)$, whose derivative has the sign of $1-a^2$ in the interior of its domain. Boundary values follow by continuity. $\square$

4. INTEGRAL ESTIMATES

The estimates in this section will be used twice: first with simple exponential majorants, and then on rational intervals. Write

$$c_m = \left(\frac{m}{m-1} \right)^{(m-1)/2}, \quad \tilde{c}_m = c_m^2, \quad d_m = \left(\frac{m}{m-2} \right)^{(m-2)/2}, \quad \tilde{d}_m = d_m^2. \quad (4.1)$$

For $m \geq 3$, one has $c_m < 5/3$, $\tilde{c}_m < 3$, $d_m < 3$, and $\tilde{d}_m < 9$.

Let $b : [0, 1] \rightarrow [0, \infty)$ be a convex majorant of β . Choose $\kappa \in [0, 1]$ with $\mathbb{E}|1-aq| \leq \kappa$, and set $\ell(t) = 1 - (1-\kappa)t$. Convexity of the modulus gives $\mathbb{E}|1-atq| \leq \ell(t)$. In particular one may take any κ with

$$\kappa^2 \geq \min\{\beta(1), Vs\}, \quad (4.2)$$

because Cauchy–Schwarz gives $\mathbb{E}|1-aq| = \mathbb{E}|\zeta q| \leq \sqrt{Vs}$.

4.1. Products with one factor omitted. The first bound uses the same Maclaurin inequality as Tao's proof of Proposition 11 [Tao26]; the bounds retaining the reciprocal second moment refine Zhang [Zha26, Lemma 3.2]. Both forms are useful, since their relative strength varies with s .

Lemma 4.1. *In the notation above, let $\mathcal{A}(t) = \sum_j |q_j|^2 \prod_{k \neq j} |1 - atq_k|$. Then*

$$\mathcal{A}(t) \leq m \min \left\{ \min\{1, c_m s\} b(t)^{(m-1)/2}, \min\{1, \tilde{c}_m s\} \ell(t)^{m-1} \right\}. \quad (4.3)$$

If $I = [l, u] \subset [0, 1]$, $a \leq a_+$, and $\delta_I = \max\{0, 1 - a_+ u\}$, then, for all $t \in I$,

$$\mathcal{A}(t) \leq ms \min \left\{ \left(\frac{mb(t) - \delta_I^2}{m-1} \right)_+^{(m-1)/2}, \left(\frac{m\ell(t) - \delta_I}{m-1} \right)_+^{m-1} \right\}. \quad (4.4)$$

Proof. Put $B_j = |1 - atq_j|$. The special case of Maclaurin's inequality $\sum_j \prod_{k \neq j} B_k \leq m(\mathbb{E}B)^{m-1}$ follows, for example, by repeatedly replacing two variables by their arithmetic mean: with their sum fixed, the elementary symmetric function is nondecreasing under this replacement. For completeness, choose a largest and a smallest coordinate at each replacement. The sum of squared deviations from the fixed mean decreases by half the square of their difference. The coordinate range therefore tends to zero, and continuity gives the equal-variable maximum. Since $|q_j|^2 \leq 1$ and $\mathbb{E}B \leq \min\{\sqrt{b(t)}, \ell(t)\}$, this proves the terms with coefficient 1 in (4.3).

For a fixed omitted index j , the arithmetic-geometric mean inequality gives $\prod_{k \neq j} B_k \leq c_m b(t)^{(m-1)/2}$ and $\prod_{k \neq j} B_k \leq \tilde{c}_m \ell(t)^{m-1}$. Multiply by $|q_j|^2$ and sum to obtain the other terms in (4.3). On I , each $B_j \geq 1 - at|q_j| \geq \delta_I$. Thus the sums over $k \neq j$ are bounded by $mb(t) - \delta_I^2$ for the squares and by $m\ell(t) - \delta_I$ for the moduli. The same argument proves (4.4). $\square$

The positive parts in (4.4) permit harmless enlargement of interval bounds. On an interval containing admissible data, the expressions before taking positive parts are already nonnegative. The lower bound δ_I is fixed throughout I ; it is not replaced by a function of t when convexity is used for integration.

4.2. Centered interpolation. Recall the centered coordinates $w_j = q_j - \mu$ from (2.4). The interpolation below is Zhang's construction [Zha26, Lemma 5.1], with the majorants b and ℓ substituted for the disk-based bounds. We also need a second estimate without a denominator, since a can exceed one and $D(t)$ can vanish.

Lemma 4.2. *For every $t \in [0, 1]$ with $D(t) \neq 0$,*

$$|F(t) - D(t)^m| \leq \frac{ma^2 t^2 v}{2|D(t)|} \min\{c_m b(t)^{(m-1)/2}, \tilde{c}_m \ell(t)^{m-1}\}. \quad (4.5)$$

For every $t \in [0, 1]$, including zeros of D ,

$$|F(t) - D(t)^m| \leq \frac{m(m-1)a^2 t^2 v}{2} \min\{d_m b(t)^{(m-2)/2}, \tilde{d}_m \ell(t)^{m-2}\}. \quad (4.6)$$

Proof. Step 1. The estimate with denominator. For $0 \leq \theta \leq 1$, define $G_\theta(t) = \prod_j (D(t) - at\theta w_j)$. Each factor is $(1 - \theta)D(t) + \theta(1 - atq_j)$. Its mean square, averaged over j , is $|D(t)|^2 + a^2 t^2 \theta^2 v \leq \beta(t)$; its mean modulus is at most $\ell(t)$, because $|D(t)| \leq \mathbb{E}|1 - atq| \leq \ell(t)$.

Using $\sum_j w_j = 0$ in the derivative of G_θ yields the polynomial identity

$$D(t) \partial_\theta G_\theta(t) = -a^2 t^2 \theta \sum_j w_j^2 \prod_{k \neq j} (D(t) - at\theta w_k).$$

The omitted-factor estimates from the proof of Lemma 4.1, followed by integration in θ , prove (4.5).

Step 2. The estimate without denominator. Since $\partial_\theta G_0(t) = 0$, Taylor's formula with integral remainder gives $G_1 - G_0 = \int_0^1 (1 - \theta) \partial_\theta^2 G_\theta d\theta$. For the ordered pairs in the second derivative, $\sum_{j \neq k} |w_j w_k| \leq m(m-1)v$ by Cauchy-Schwarz. The products with two factors omitted are bounded by $d_m b^{(m-2)/2}$ and $\tilde{d}_m \ell^{m-2}$. This proves (4.6). No division by D or v is used in this argument. $\square$

Suppose $\mathcal{C}$ is any upper bound for $|hJ|$, and $\mathcal{U}(t)$ is any majorant satisfying $|F(t) - D(t)^m| \leq a^2 t^2 v \mathcal{U}(t)$. Lemmas 3.2 and 3.1 imply the necessary inequalities

$$1 \leq \kappa^m + \frac{m}{m+1} \mathcal{C} + a^2 \int_0^1 t \mathcal{A}(t) dt, \quad (4.7)$$

$$1 \leq \mathcal{C} + (Vv)^{(m+1)/2} + Kv, \quad K = (m+1)a^3 r \int_0^1 t^2 \mathcal{U}(t) dt. \quad (4.8)$$

There is also a useful version which does not use $\mathcal{C}$. Since the *actual* quantity $|hJ|$ is at most r by (3.8), division by $1 - r > 0$ in (3.7) gives

$$1 \leq (1+r) \left\{ V^{(m+1)/2} (1-r^2)^{(m-1)/2} + K \right\}. \quad (4.9)$$

Here we used $v \leq 1 - r^2$. An arbitrarily chosen interval upper bound $\mathcal{C}$ need not itself be at most r ; (4.9) uses (3.8) before interval enlargement.

4.3. Convex integration on a rational mesh. For a nonnegative convex function f on $[l, u]$, its endpoint chord gives

$$\int_l^u t f(t) dt \leq \frac{(u-l)(2l+u)}{6} f(l) + \frac{(u-l)(l+2u)}{6} f(u), \quad (4.10)$$

$$\int_l^u t^2 f(t) dt \leq \frac{(u-l)(3l^2 + 2lu + u^2)}{12} f(l) + \frac{(u-l)(l^2 + 2lu + 3u^2)}{12} f(u). \quad (4.11)$$

These formulas follow by integrating the affine chord against t and t^2 ; all four weights are nonnegative.

Every separate power of b or ℓ in Lemma 4.1 is convex when $m \geq 3$. The same holds for the bounds in Lemma 4.2 when $m \geq 4$. On a mesh interval where $|D|$ has a positive constant lower bound, that constant is used in (4.5). If there is no such lower bound, only (4.6) is used. We integrate each available convex majorant separately by (4.10) or (4.11) and then take the smallest resulting upper bound. We do not assert that the pointwise minimum of convex functions is convex. For $m = 3$, the exponent $(m-2)/2$ is below one; the centered integral tests are therefore not used in that degree.

5. THE ANALYTIC ARGUMENT FOR LARGE DEGREES

The covariance and integral bounds exclude large degrees without finite verification. We use a fixed cutoff that leaves comfortable margins in each of the four ranges of a below.

Proposition 5.1. *There is no normalized counterexample with $m \geq 100000$.*

Proof. Throughout this proof we may replace V_m by 1. From (3.1) and (3.2),

$$x \geq a(1 - \sqrt{1-s}) \geq a \left(1 - \frac{a}{\sqrt{1+a^2}} \right) \geq \frac{a}{2(1+a^2)}. \quad (5.1)$$

The last step follows by rationalizing the difference. Define $g(a) = a^2/[4(1+a^2)^2]$. Since $\rho \geq r^2 \geq x^2$, (2.5) and (5.1) give $\beta(t) \leq 1 - g(a)t$. More generally, whenever $\beta(t) \leq 1 - gt$ with $g > 0$, Lemma 4.1 and (3.4) give

$$|F(1)| \leq e^{-mg/2}, \quad |\mathcal{R}| \leq \frac{4ma^2}{(m-1)^2 g^2}. \quad (5.2)$$

Indeed, use $\mathcal{A}(t) \leq m\beta(t)^{(m-1)/2}$ and extend the integral of $te^{-(m-1)gt/2}$ to $[0, \infty)$. From (3.6), (3.5), and (3.8), we have

$$1 \leq |F(1)| + \psi(a) + |\mathcal{R}|. \quad (5.3)$$

We now consider four ranges of a .

Step 1. $0 < a \leq 1/10$. The strict inequalities $|a - \zeta_j| > 1$ imply $|P'(a)| > m + 1$. The arithmetic-geometric mean bound used in Lemma 3.3 yields $m \log(1 + a^2) > 2 \log(m + 1) - m \log(1 + 1/m) > 2 \log(m + 1) - 1$. Consequently

$$ma^2 > 2 \log(m + 1) - 1 > 19. \quad (5.4)$$

The middle expression in (5.1) is at least $9a/10$ in this range. Hence $\beta(t) \leq 1 - (4/5)a^2t$. In (5.2) take $g = (4/5)a^2$. Using (5.4), we obtain $|F(1)| < 1/100$ and $|\mathcal{R}| < 8/(ma^2) < 8/19$. Also $\psi(a) < 1/6$. The right side of (5.3) is therefore less than $1/100 + 1/6 + 8/19 < 1$.

Step 2. $a \in [1/10, 3/4] \cup [5/4, 2]$. Here $g(a) \geq 1/500$, so $|F(1)| < 1/1000$. Set $H_0(a) = 16(1 + a^2)^4/a^2$. Equation (5.2) gives $|\mathcal{R}| < 5H_0(a)/m$. The endpoint bounds are listed in Table 1. To check the table, ψ increases before 1 and decreases afterwards, while the

TABLE 1. Bounds away from $a = 1$ in the large-degree argument.

Range of a	Upper bound for $\psi(a)$	Upper bound for $H_0(a)$
$[1/10, 1/2]$	$3/4$	1665
$[1/2, 3/4]$	$19/20$	170
$[5/4, 3/2]$	$19/20$	794
$[3/2, 2]$	$17/20$	2500

logarithmic derivative of $(1 + u)^4/u$ has the sign of $3u - 1$. Thus only the displayed endpoints need be checked. The exponential comparisons are $e^{3/8} < 3/2$, $e^{7/32} < 19/15$, $e^{9/32} > 25/19$, and $e^{5/8} > 30/17$; they follow by the exponential series (an exact check is included in the supplement). In all four rows the sum of the last two bounds in (5.3), with the factor $5/m$ on H_0 , and $1/1000$ is less than one. The largest such sum is $9907/10000$.

Step 3. $3/4 \leq a \leq 5/4$. Equation (5.1) gives $r \geq 6/25$. Put $B_* = 589/625$. Then $\rho \geq 36/625$, $\beta(1) \leq B_*$, and $a^2s \geq 9/25$ by (3.2). Thus

$$\beta(t) \leq 1 - t/5 \quad (0 \leq t \leq 1/2), \quad \beta(t) \leq B_* \quad (1/2 \leq t \leq 1). \quad (5.5)$$

For the second inequality use convexity, $\beta(1/2) \leq 9/10$, and the endpoint bound at 1. Also $|D(t)| \geq 3/8$ on $[0, 1/2]$.

Let $E = (m + 1)|h| \int_0^1 |F - D^m| dt$, and split it as $E_1 + E_2$ at $1/2$. Lemma 4.2 and (5.5) imply

$$E_1 \leq \frac{78125}{9} \frac{(m + 1)m}{(m - 1)^3} v \leq \frac{10000}{m} v \leq \frac{v}{10}, \quad (5.6)$$

$$E_2 \leq 2(m + 1)m^2 v B_*^{(m-2)/2} \leq 2(m + 1)m^2 v e^{-m/40} \leq \frac{6 \cdot 10^{12}}{m^3} v < \frac{v}{10}. \quad (5.7)$$

For (5.6), use $c_m < 5/3$, $a \leq 5/4$, $|h| \leq 5/4$, and integrate $t^2 e^{-(m-1)t/10}$ on $[0, \infty)$. For (5.7), use the nonsingular estimate, $d_m < 3$, and $\int_{1/2}^1 t^2 dt = 7/24$; the final inequality follows from $e^{m/40} \geq (m/40)^6/720$. Since $v \leq 1 - r^2 \leq B_*$, $v^{(m+1)/2} \leq 2(1 - r)B_*^{(m-1)/2} \leq (1 - r)/5$. Now (3.7), (3.2), (3.8), (5.6), and (5.7) give

$$1 \leq r + \frac{1 - r}{5} + \frac{v}{5} \leq r + \frac{3}{5}(1 - r) < 1.$$

Step 4. $a \geq 2$. From (3.1), $a^2 s \leq 2ar - r^2 \leq 2a - 1$, so $s \leq 3/4$. Moreover $\beta(1) \leq 1$ and Minkowski's inequality give $a\sqrt{s} \leq 1 + \sqrt{\beta(1)} \leq 2$. Therefore $a^2 s \leq 4$ and $\mathbb{E}|1 - aq| \leq \sqrt{s} \leq \sqrt{3}/2 < 7/8$. Use $\ell(t) = 1 - t/8$ in Lemma 4.1. Then $|F(1)| \leq (7/8)^m$ and

$$|\mathcal{R}| \leq 12m \int_0^8 t(1 - t/8)^{m-1} dt = \frac{768}{m+1}.$$

Also $\psi(a) \leq 2e^{-3/2} < 9/20$. These bounds contradict (5.3). The four steps exhaust all $a > 0$. $\square$

6. PRODUCT BOUNDS FOR THE SMALL DEGREES

The direct estimates alone lose information when both the reciprocal mean and the relevant products are close to their extremal values. We next retain the covariance of the original root distances and the variation of the reciprocal moduli. All notation from (2.2)–(2.4) remains in force.

6.1. An elementary symmetric inequality. Recall the elementary symmetric polynomials e_k from Section 2. The following estimate gives the improvement over the arithmetic–geometric mean inequality needed in Lemma 6.3.

Lemma 6.1. *If $m \geq 3$, $Y_j \geq 0$, and $\sum_j Y_j = m$, then*

$$e_{m-1}(Y) \leq 3 + (m-3)e_m(Y). \quad (6.1)$$

Proof. Step 1. Boundary values. Maximize $f = e_{m-1} - (m-3)e_m$ on the closed simplex. On its boundary, $f = 0$ if two or more coordinates vanish. If exactly one vanishes, the arithmetic–geometric mean inequality gives $f \leq (m/(m-1))^{m-1} < 3$. At the point $(1, \dots, 1)$, $f = 3$.

Step 2. Interior stationary points. It remains to exclude a nonconstant interior maximum. At an interior stationary point, two unequal coordinates Y_i, Y_j satisfy

$$\sum_{\ell \neq i, j} \frac{1}{Y_\ell} = m - 3. \quad (6.2)$$

Indeed, the difference of the corresponding partial derivatives is $(Y_j - Y_i) \prod_{\ell \neq i, j} Y_\ell (\sum_{\ell \neq i, j} Y_\ell^{-1} - (m-3))$. If three distinct values occurred, applying (6.2) to two pairs sharing an index would make the other two coordinates equal. Thus at most two values, say $u \neq w$, occur.

Step 3. Exclusion of a nonconstant interior maximum. Suppose both occur at least twice. For a pair of equal coordinates u, u , the second derivative along the direction $(1, -1)$ is nonpositive only if $\sum_{\ell \neq i, j} Y_\ell^{-1} - (m-3) \geq 0$. Subtracting (6.2) for a mixed pair u, w gives $1/w - 1/u \geq 0$, hence $u \geq w$. The pair w, w gives $w \geq u$, a contradiction. Therefore one value, say u , occurs $m-1$ times. Equation (6.2) becomes $(m-2)/u = m-3$. For $m=3$ this is impossible; for $m>3$ it forces the remaining coordinate to be $m - (m-1)(m-2)/(m-3) = -2/(m-3) < 0$. There is no nonconstant interior maximum, which proves (6.1). $\square$

6.2. The original root distances. Put $u_j = a - z_j$, where the z_j are the zeros in (2.2), and define

$$\begin{aligned} A_0 &= \mathbb{E}u = \frac{(m+1)a}{m}, & B_0 &= \mathbb{E}|u|^2 = \frac{(m+1)(1+a^2)}{m}, \\ C_0 &= B_0 - A_0^2 = \frac{(m+1)(m-a^2)}{m^2}. \end{aligned} \quad (6.3)$$

The logarithmic derivative identity of Meir and Sharma [MS69, equation (3.2)] relates these root distances to the reciprocal critical-point mean. Tao [Tao26, Lemma 15] states its simple-zero case explicitly.

Lemma 6.2 ([MS69, equation (3.2)]; [Tao26, Lemma 15]). *For the original root distances $u_j = a - z_j$ in (6.3), one has*

$$\mathbb{E} \frac{1}{u} = \frac{\mu}{2}. \quad (6.4)$$

Lemma 6.3. *If $C_0 > 0$, then*

$$Q_0 \geq \frac{(m+1)^2}{B_0^m} (1 + \gamma), \quad \gamma = \frac{m}{3} \frac{|A_0 - B_0 \mu/2|^2}{C_0}. \quad (6.5)$$

If $C_0 = 0$, then $A_0 - B_0 \mu/2 = 0$ and the same bound holds with $\gamma = 0$.

Proof. Put $T = \mathbb{E}|1/u|^2$ and $M = \mathbb{E}(1/u) = \mu/2$, where the last equality is Lemma 6.2. Cauchy–Schwarz applied to $u - A_0$ and $1/u - M$ gives $|1 - A_0 M|^2 \leq C_0(T - |M|^2)$. Rearranging yields

$$B_0 T \geq 1 + \frac{|A_0 - B_0 M|^2}{C_0}. \quad (6.6)$$

Apply Lemma 6.1 to $Y_j = |u_j|^2/B_0$. Since $e_{m-1}(Y) = e_m(Y) m B_0 T$, equations (6.1) and (6.6) imply $e_m(Y)(1 + \gamma) \leq 1$. Finally, $e_m(Y) = (m+1)^2/(B_0^m Q_0)$, proving (6.5). If $C_0 = 0$, all $u_j = A_0$, and (6.4) gives $M = 1/A_0$. The asserted vanishing and the weaker product bound follow directly. $\square$

6.3. Variation of the reciprocal moduli. Write $X_j = |q_j|^2$, so that $0 < X_j < 1$, $\mathbb{E}X = s$, and $Q_0 = \prod_j X_j$.

Lemma 6.4. *If $V > 0$, then*

$$Q_0 \leq s^m \exp \left(-\frac{m|\mu - as|^2}{2sV} \right). \quad (6.7)$$

For $V = 0$, the bound $Q_0 \leq s^m$ remains valid.

Proof. For $0 < u, s \leq 1$,

$$\log u \leq \log s + \frac{u - s}{s} - \frac{(u - s)^2}{2s}. \quad (6.8)$$

To verify this, the derivative with respect to u of the difference between the left and right sides is $(u - 1)(u - s)/(su)$, and that difference vanishes at $u = s$. Summing (6.8) with $u = X_j$ gives $\log Q_0 \leq m \log s - m \text{Var}(X)/(2s)$. The identity $\mu - as = -\mathbb{E}[\bar{\zeta}(X - s)]$ and Cauchy–Schwarz give $|\mu - as|^2 \leq V \text{Var}(X)$. This proves (6.7). $\square$

We next use the following product inequality from Tao’s exposition. Its nonvanishing hypothesis is automatic for reciprocal critical-point coordinates.

Lemma 6.5 ([Tao26, Lemma 13]). *If $0 < |\xi_j| \leq 1$ for $1 \leq j \leq N$, then*

$$\left(\prod_{j=1}^N |\xi_j| \right) \sum_{j=1}^N \left| \frac{1}{\xi_j} - \bar{\xi}_j \right| \leq 1 - \prod_{j=1}^N |\xi_j|^2.$$

This inequality and the reciprocal second moment give two bounds on $|a - \mu|$.

Lemma 6.6. *Put $\Delta = 1 - s$. If $m\Delta < 1$, then*

$$|a - \mu|^2 \leq \frac{\Delta^2}{1 - m\Delta}, \quad V_* \leq 1 + \frac{\Delta}{1 - m\Delta} - a^2. \quad (6.9)$$

Without this additional restriction,

$$m|a - \mu| \leq Q_0^{-1/2} - Q_0^{1/2}. \quad (6.10)$$

Proof. Since $\sum_j(1 - X_j) = m\Delta$, each $X_j \geq 1 - m\Delta$. Also $\mathbb{E}(1/q) = a$, by (2.3). Hence $|a - \mu| = |a - \bar{\mu}| \leq \mathbb{E}[(1 - X)/\sqrt{X}]$ and $V_* = \mathbb{E}(1/X) - a^2$. These identities imply (6.9).

For (6.10), apply Lemma 6.5 with $N = m$ and $\xi_j = q_j$. Since $\sum_j(1/q_j - \bar{q}_j) = m(a - \bar{\mu})$ and $\prod_j |q_j| = \sqrt{Q_0}$, the triangle inequality gives the result. $\square$

Meir and Sharma [MS69, Lemma 2] maximize an inverse-power sum under bounds on the individual variables and their product. The same extremal configuration gives the following two estimates. We give the convexity argument for both functions that occur here.

Lemma 6.7. *Suppose $0 < L \leq Q_0$, $s \leq s_+$, and $0 < \eta < s_+ \leq 1$ satisfy*

$$\eta \left(\frac{ms_+ - \eta}{m - 1} \right)^{m-1} < L. \quad (6.11)$$

Then $X_j > \eta$ for all j . Assume also $L \leq 1$, and choose an integer $k \in \{0, \dots, m - 1\}$ such that $\eta^{k+1} \leq L \leq \eta^k$. Put $z = L/\eta^k$. Then

$$m|a - \mu| \leq M := \frac{k(1 - \eta)}{\sqrt{\eta}} + \frac{1 - z}{\sqrt{z}}, \quad (6.12)$$

$$V_* \leq T - a^2, \quad T := \frac{k/\eta + 1/z + m - k - 1}{m}. \quad (6.13)$$

Proof. If $X_j = x \leq \eta$, the arithmetic–geometric mean inequality gives $Q_0 \leq x[(ms_+ - x)/(m - 1)]^{m-1}$. This expression increases for $0 \leq x \leq s_+$, contradicting (6.11). Notice that (6.11) also implies $L > \eta^m$, so the specified k exists.

Set $t_j = -\log X_j$ and $b = -\log \eta$. Then $0 \leq t_j \leq b$ and $\sum_j t_j \leq -\log L$. For any convex nondecreasing function f on $[0, b]$, moving mass from the smaller of two interior coordinates to the larger, with their sum fixed, does not decrease $\sum_j f(t_j)$. Repetition leaves at most one interior coordinate. Increasing the total mass up to $-\log L$ is permitted because $-\log L < mb$. The resulting vector has k coordinates b , one coordinate $-\log z$, and all remaining coordinates zero.

Apply this argument first to $f(t) = 2 \sinh(t/2)$ and then to $f(t) = e^t$. The identities $\mathbb{E}(1/q) = a$ and $V_* = \mathbb{E}(1/X) - a^2$ give $m|a - \mu| \leq \sum_j 2 \sinh(t_j/2)$ and $m(V_* + a^2) = \sum_j e^{t_j}$. Substituting the extremal vector proves (6.12) and (6.13). $\square$

7. CENTERED COEFFICIENTS AND CONFIGURATIONS NEAR EQUALITY

The centered interpolation controls the full product by its variance. For small degrees, it is useful to keep its coefficients separately. Throughout this section, $w_j = q_j - \mu$ and $v = \mathbb{E}|w|^2$ are the quantities defined in (2.4). Repeated application of Schoenberg's inequality gives the coefficient bound we need.

Lemma 7.1. *For $2 \leq k \leq m$,*

$$|e_k(w)| \leq \binom{m}{k} \left(\frac{k-1}{m-1} v \right)^{k/2}. \quad (7.1)$$

Proof. Let $R(z) = \prod_j(z - w_j)$. The centroid of the zeros of every derivative of R is zero. Apply Lemma 2.1 successively to $R, R', \dots, R^{(m-k-1)}$, using degrees $m, m-1, \dots, k+1$. If $\lambda_1, \dots, \lambda_k$ are the zeros of $R^{(m-k)}$, then $\sum_j |\lambda_j|^2 \leq k(k-1)v/(m-1)$. The constant term of the monic normalization of $R^{(m-k)}$ is $(-1)^k e_k(w)/\binom{m}{k}$. Thus (7.1) follows from the arithmetic–geometric mean inequality for $|\lambda_j|^2$. For $k = m$, the same argument uses $\sum_j |w_j|^2 = mv$ directly. $\square$

For nonnegative integers k, p and $d \geq 0$, define

$$S_{k,p}(d) = \sum_{j=0}^p \binom{k+j}{k} d^j, \quad I_{k,p}(d) = \frac{S_{k,p}(d)}{(k+p+1)\binom{k+p}{k}} = \int_0^1 t^k (1-t+dt)^p dt. \quad (7.2)$$

The last equality follows by expanding $(1 - t + dt)^p$ and integrating each beta integral.

Lemma 7.2. *If $\mathcal{C} \geq |hJ|$ and $d \geq |1 - h|$, then every normalized counterexample satisfies*

$$1 \leq \mathcal{C} + d^{m+1} + ar \sum_{k=2}^m \left(\frac{a^2 v(k-1)}{m-1} \right)^{k/2} S_{k,m-k}(d). \quad (7.3)$$

Proof. Since $e_1(w) = 0$,

$$F(t) = D(t)^m + \sum_{k=2}^m (-at)^k e_k(w) D(t)^{m-k}. \quad (7.4)$$

The bound $|D(t)| \leq 1 - t + dt$, Lemma 7.1, and (7.2) apply term by term in (3.7). Since $(m+1)\binom{m}{k}I_{k,m-k}(d) = S_{k,m-k}(d)$, this gives (7.3). $\square$

7.1. Retaining the quadratic term. When $r > 0$, write $\mu = re^{i\theta}$ and put $q'_j = e^{-i\theta}q_j$, $w'_j = q'_j - r$. Then $\mathbb{E}q' = r$. Let $W = \text{Var}(\text{Re } q')$. The identity $e_2(w') = -\frac{1}{2} \sum_j (w'_j)^2$ gives

$$\text{Re } e_2(w') = \frac{m}{2}v - mW. \quad (7.5)$$

To estimate W , we record the classical variance bound for a probability distribution on an interval.

Lemma 7.3 ([BD00, p. 353]). *Let $Y_1, \dots, Y_N \in [L, U]$ be real numbers, and let $\omega_1, \dots, \omega_N \geq 0$ satisfy $\sum_j \omega_j = 1$. If $b = \sum_j \omega_j Y_j$, then $\sum_j \omega_j (Y_j - b)^2 \leq (U - b)(b - L)$.*

For the reciprocal data, the interval endpoints can be bounded explicitly.

Lemma 7.4. *Suppose $r > 0$ and $|q_j|^2 \geq t_0 \geq 0$ for all j . Set*

$$L_r = \max \left\{ -1, mr - (m-1), \frac{m^2 r^2 + t_0 - (m-1)^2}{2mr} \right\}. \quad (7.6)$$

Then

$$\text{Re } q'_j \geq L_r, \quad W \leq \min\{(m-1)(1-r)^2, (1-r)(r - L_r)\}. \quad (7.7)$$

Proof. The bounds $|q'_j| \leq 1$ and $\sum_j q'_j = mr$ give the first two entries in (7.6). Also $|mr - q'_j| \leq m-1$; squaring and using $|q'_j|^2 \geq t_0$ gives the third. Apply Lemma 7.3 with $N = m$, $Y_j = \text{Re } q'_j$, $\omega_j = 1/m$, $L = L_r$, $U = 1$, and $b = r$ to obtain the second bound on W . For the first, let $d_j = 1 - \text{Re } q'_j \geq 0$. Since $\sum_j d_j = m(1-r)$, $W = m^{-1} \sum_j d_j^2 - (1-r)^2 \leq (m-1)(1-r)^2$. $\square$

Define

$$K_2(h) = (m+1)h \int_0^1 t^2 (1 - ht)^{m-2} dt. \quad (7.8)$$

Its contribution to the last term of (3.7) is $-a^2 e_2(w) K_2(h)$. We shall keep its real part instead of replacing it by an absolute-value error.

Lemma 7.5. *Suppose $0 < h_0 := ar \leq 1$, $\varepsilon \geq |e^{i\theta} - 1|$, and $d \geq |1 - h|$. Then*

$$|e^{2i\theta} K_2(h) - K_2(h_0)| \leq E_2 := (m+1)h_0 \varepsilon \{3I_{2,m-2}(d) + (m-2)h_0 I_{3,m-3}(d)\}. \quad (7.9)$$

If $0 < L_2 \leq \text{Re } e_2(w')$, $0 \leq K_- \leq K_2(h_0)$, and $\mathcal{C} \geq |hJ|$, then

$$\begin{aligned} 1 \leq \mathcal{C} + d^{m+1} - a^2 K_- L_2 + \frac{ma^2 v}{2} E_2 \\ + ar \sum_{k=3}^m \left(\frac{a^2 v(k-1)}{m-1} \right)^{k/2} S_{k,m-k}(d). \end{aligned} \quad (7.10)$$

Proof. We have $h = h_0 e^{i\theta}$ and $0 \leq 1 - h_0 t \leq |1 - ht| \leq 1 - t + dt$. In the difference of the integrands defining $e^{2i\theta} K_2(h)$ and $K_2(h_0)$, first estimate $|e^{3i\theta} - 1| \leq 3\varepsilon$, and then telescope the difference of the $(m-2)$ nd powers. This gives (7.9) by (7.2).

Since $e_2(w) = e^{2i\theta} e_2(w')$ and $|e_2(w')| \leq mv/2$ by Lemma 7.1, $\operatorname{Re}(e_2(w)K_2(h)) \geq K_-L_2 - (mv/2)E_2$. Take real parts in (3.7), use (7.4) for the quadratic term, and estimate the remaining terms as in Lemma 7.2. This proves (7.10). $\square$

7.2. An analytic neighborhood of $r = 1$. The finite verification must not leave an unchecked neighborhood of a limiting equality configuration. The next lemma treats that region uniformly, without computation.

Lemma 7.6. *There is no normalized counterexample with $3 \leq m \leq 12$ and $r \geq 99999/100000$.*

Proof. Put $\delta = 1 - r$. Then $0 < \delta \leq 10^{-5}$, and $\Delta = 1 - s \leq 2\delta$.

Step 1. Control of the phase. Each $X_j = |q_j|^2 \geq 1 - m\Delta \geq 1 - 24\delta > (99/100)^2$. Lemma 6.6 gives $|a - \mu| \leq 2\delta/\sqrt{1 - 24\delta} < (21/10)\delta$. It follows that

$$1 - 4\delta \leq a \leq 1 + 2\delta, \quad a^2 < 2, \quad |e^{i\theta} - 1| < 5\delta, \quad |h - 1| < 12\delta. \quad (7.11)$$

For the phase bound, use $|a - r| \leq |a - \mu|$ and $|\mu - r| \leq |\mu - a| + |a - r| < (21/5)\delta$, then divide by r . The bound for $h - 1$ follows from $|a\mu - 1| \leq a|\mu - r| + |ar - 1|$. Also $|D(t)| \leq 1 - t + 12\delta t \leq 1$ on $[0, 1]$.

Step 2. The quadratic contribution. From (7.8), $K_2(1) = 2/[m(m-1)]$. Telescoping the difference of the powers and using $|D(t)| \leq 1$ gives

$$|K_2(h) - K_2(1)| \leq (m+1)|h - 1| \left(\frac{1}{3} + \frac{m-2}{4} \right) \leq 442\delta.$$

Using (7.11) once more, we obtain $|e^{2i\theta} K_2(h) - K_2(1)| < 450\delta$. Equations (7.5) and (7.7), together with $|e_2(w')| \leq mv/2$, therefore imply

$$\operatorname{Re}(e_2(w)K_2(h)) \geq \frac{v}{m-1} - 2\delta^2 - 225m\delta v \geq -2\delta^2. \quad (7.12)$$

The last step uses $m \leq 12$ and $\delta \leq 10^{-5}$.

Step 3. The higher coefficients. Set $\tau = a\sqrt{v}$. Since $v \leq 1 - r^2 \leq 2\delta$, we have $\tau \leq 2\sqrt{2\delta} < 1/100$. The binomial series gives $S_{k,m-k}(12\delta) \leq (1 - 12\delta)^{-k-1} < 2$ for $3 \leq k \leq m \leq 12$. By (7.11), $|h| < 2$. Lemma 7.1 and the calculation in Lemma 7.2 thus bound the sum of the terms with $k \geq 3$ by $4 \sum_{k=3}^{\infty} \tau^k < 5\tau^3 < 128\delta^{3/2}$.

Now use (3.8), (3.7), and (7.12). The quadratic term is at most $4\delta^2$, the endpoint term is at most $(12\delta)^4$, and hence

$$1 \leq 1 - \delta + 4\delta^2 + 128\delta^{3/2} + (12\delta)^4.$$

This is impossible, because $4\delta + 128\sqrt{\delta} + 20736\delta^3 < 4/10^5 + 128/300 + 20736/10^{15} < 1/2$. $\square$

8. FINITE VERIFICATION AND COMPLETION OF THE PROOF

We first reduce the root position to a fixed bounded interval. This reduction is independent of the cutoff in Proposition 5.1.

Lemma 8.1. *Every normalized counterexample satisfies $0 < a < 5$.*

Proof. Suppose $a \geq 5$. From (3.1) and (3.2), $a^2 s \leq 2ar$ and $ar \leq 1 + \sqrt{s}$. Hence, with $s_0 = \sqrt{s}$, one has $25s_0^2 \leq 2 + 2s_0$, which implies $s_0 < 1/3$ and $a^2 s < 8/3$. We may therefore use $\ell(t) = 1 - 2t/3$ in Lemma 4.1. It gives $|F(1)| \leq 3^{-m}$ and $|\mathcal{R}| \leq 8m \int_0^{3/2} t(1 - 2t/3)^{m-1} dt = 18/(m+1)$. Also (3.8) gives $|hJ| \leq rs^{m/2} \leq 3^{-(m+1)}$. Since $a^2 \leq m$, we have $m+1 \geq 26$. Equations (3.5) and (3.6) would then imply $1 \leq 3^{-m} + 3^{-(m+1)} + 18/(m+1) \leq 3^{-25} + 3^{-26} + 9/13 < 1$, a contradiction. $\square$

Thus every normalized counterexample gives a point of

$$(a, r, s) \in [0, 5] \times [0, 1] \times [0, 1]. \quad (8.1)$$

The endpoints have been included deliberately. This permits the use of closed dyadic boxes; their common boundaries are checked along with their interiors.

Proposition 8.2. *There is no normalized counterexample with $3 \leq m \leq 99999$.*

Proof. The files `certificate_n4_to_n13.json` and `certificate_n14_to_n100000.json` in the supplement describe finite binary partitions of (8.1). Each internal node bisects one coordinate interval at its exact midpoint. Each leaf specifies one of the inequalities in Appendix A. The checkers reconstruct the boxes from their paths, verify that every internal node has both children, and evaluate the designated inequality on the entire leaf. They also verify that the degree blocks are consecutive, with first degree 4 and last degree 100000.

For every leaf, either a necessary inequality from Lemmas 3.1, 3.3, and 6.3–6.7 fails strictly; or Lemma 7.6 applies; or one of (4.7), (4.8), (4.9), (7.3), and (7.10) has a right side strictly below one. The parameter bounds used for these tests, including the direction of every degree-dependent estimate, are specified in Appendix A. The arithmetic in Appendix B gives rigorous upper bounds, so each accepted leaf excludes all normalized counterexamples in that box.

The complete computation at scale 2^{100} gives the following counts. The reported margins concern the tests whose final upper bound is compared with one; the elementary rational contradictions and Lemma 7.6 are checked separately.

Degrees n	Degree blocks	Leaf boxes	Maximum depth
4–13	10	12821	53
14–100000	62	20850	33
Total	72	33671	

Writing the final upper bounds as integers divided by $S = 2^{100}$, the smallest positive margins $S - U$ are respectively

$$859210771742754798767913 \quad \text{and} \quad 3409997860291474113416360. \quad (8.2)$$

Both full verifications also pass at scale 2^{160} . The exact logs, file hashes, and commands are supplied in Appendix B and the supplement. In particular, (8.2) records strictly positive separation from the acceptance threshold for these tests. Since the reconstructed partitions cover (8.1), Lemma 8.1 completes the proof. $\square$

We now prove Theorem 1.2.

Proof. If $\sigma_2(P) = 0$, all zeros coincide and (1.3) is immediate. A multiple zero is itself a critical point. Degrees two and three follow from Lemma 2.2. Any remaining failure of (1.3) can be normalized as in (2.2). Equations (2.3) exclude $a = 0$. For $a > 0$, Proposition 8.2 excludes $4 \leq n \leq 100000$, and Proposition 5.1 excludes $n \geq 100001$. For sharpness, take $P(z) = z^n - 1$. Its centroid is zero, its standard deviation in (1.2) is one, and its only critical point is zero. Every zero is at distance one from that critical point. $\square$

APPENDIX A. BOUNDS ON A PARAMETER BOX

This appendix gives the mathematical specification of the finite verification. The labels of tests in the data files are only program identifiers; they do not introduce additional mathematical assertions. All interval operations below are interpreted as implications under the assumption that the box contains a normalized counterexample.

Write a box as $\mathcal{B} = [a_-, a_+] \times [r_-, r_+] \times [s_-, s_+]$. Define

$$\begin{aligned} v_+ &= s_+ - r_-^2, & v_- &= (s_- - r_+^2)_+, & d_a &= \max\{0, a_- - r_+, r_- - a_+\}, \\ d_h &= \max\{0, 1 - a_+r_+, a_-r_- - 1\}, & \Delta_+ &= 1 - s_-. \end{aligned}$$

Thus $v \leq v_+$, $v \geq v_-$, $|a - \mu| \geq d_a$, and $|1 - a\mu| \geq d_h$.

A.1. Refinements for a fixed small degree. In this subsection $m \in \{3, \dots, 12\}$ is fixed. Begin with $V_0 = (m - 1)/m$. If $t_* = 1 - m\Delta_+ > 0$, set

$$D_+^2 = \frac{\Delta_+^2}{t_*}, \quad V_0 = \min \left\{ \frac{m-1}{m}, 1 + \frac{\Delta_+}{t_*} - a_-^2 \right\}.$$

Otherwise D_+^2 is initially unspecified. Lemma 6.6 shows that, when specified, $|a - \mu|^2 \leq D_+^2$, and in all cases $V_* \leq V_0$. A negative value of V_0 excludes the box immediately.

For product estimates define

$$\begin{aligned} B_+ &= \frac{m+1}{m}(1 + a_+^2), & C_+ &= \frac{m+1}{m^2}(m - a_-^2)_+, \\ g_- &= \max \left\{ 0, a_- - \frac{1 + a_+^2}{2}r_+, \frac{1 + a_-^2}{2}r_- - a_+ \right\}, \\ \gamma_- &= \begin{cases} \frac{m}{3} \left(\frac{m+1}{m} \right)^2 \frac{g_-^2}{C_+}, & C_+ > 0, \\ 0, & C_+ = 0, \end{cases} & L &= \frac{(m+1)^2(1 + \gamma_-)}{B_+^m}. \end{aligned}$$

Equation (6.3) and Lemma 6.3 give $L \leq Q_0$. The case $C_0 = 0$ causes no division by zero: its covariance gap vanishes, so a positive lower bound g_- cannot occur in a box containing such data.

Let $d_s = \max\{0, a_-s_- - r_+, r_- - a_+s_+\}$. If $s_+V_0 > 0$, put $\lambda_- = md_s^2/(2s_+V_0)$; otherwise put $\lambda_- = 0$. Lemma 6.4 gives

$$Q_0 \leq s_+^m e^{-\lambda_-}. \quad (\text{A.1})$$

The value $\lambda_- = 0$ when $V_0 = 0$ uses only the arithmetic-geometric mean bound, so it requires no division by V_0 .

If $0 < L \leq s_+^m$, choose η by 30 exact bisections on $[0, s_+]$, retaining a lower endpoint for which (6.11) holds with s_+ in place of s . Whenever this lower endpoint is positive, Lemma 6.7 supplies k, z, M, T . An upward-rounded upper bound $M_+ \geq M$ is permitted. Refine

$$D_+^2 \leftarrow \min\{D_+^2, (M_+/m)^2\}, \quad V = \min\{V_0, T - a_-^2\}; \quad (\text{A.2})$$

the first minimum ignores an initially unspecified D_+^2 . If this refinement is unavailable, use $V = V_0$. The product loss in (A.1) is kept with its original V_0 , so the order of these refinements introduces no circularity.

For clarity, Table 2 lists all elementary exclusion tests. The first six tests may use V_0 ; the tests `moment_A_packed` and `moment_B_packed` use the refined value V . Every displayed strict inequality in that table implies that the box contains no normalized counterexample.

Here $T_{32}(x) = \sum_{j=0}^{32} x^j/j! \leq e^x$ for $x \geq 0$. The function $(1 - L)/\sqrt{L}$ decreases on $(0, 1]$, which justifies the test `product_radial` by (6.10). The test `finite_radial` can hold only if $1 - m\Delta_+ > 0$. The last row is precisely Lemma 7.6; it is not a numerical approximation to that lemma.

For the remaining tests assume $V \geq 0$ and $v_+ \geq 0$, and set

$$H = \max\{r_-^2 + (1 - V)(1 - s_+), 1 - (a_+^2 + V)v_+\}, \quad A = a_-^2 s_-.$$

If D_+^2 is available, include $a_-^2(1 - s_+) + r_-^2 - D_+^2$ as a third entry in the maximum defining H . Lemma 3.1 then gives $\rho \geq H$ and $\beta(t) \leq b(t) := 1 - Ht - At(1 - t)$. The extra entry follows from the identity $\rho = a^2(1 - s) + r^2 - |a - \mu|^2$. If $H > 1$, or if the interior minimum of b is negative, the box is excluded; this is `beta_negative`. Otherwise the checker requires

TABLE 2. Elementary exclusion tests for a fixed degree.

Identifier	Sufficient condition for exclusion
<code>degree_bound</code>	$a_-^2 > m$
<code>variance_nonnegative</code>	$v_+ < 0$
<code>inverse_moment</code>	$(a_+^2 + V_0)s_+ < 1$
<code>moment_A</code>	$d_a^2 > (a_+^2 + V_0 - 1)_+(1 - s_-)$
<code>moment_B</code>	$d_h^2 > V_0 v_+$
<code>product</code>	$(B_+ s_+)^m < (m + 1)^2$
<code>finite_radial</code>	$d_a^2(1 - m\Delta_+) > \Delta_+^2$
<code>product_stable</code>	$(B_+ s_+)^m < (m + 1)^2 T_{32}(\lambda_-)$
<code>product_root_stable</code>	$(B_+ s_+)^m < (m + 1)^2 (1 + \gamma_-) T_{32}(\lambda_-)$
<code>product_radial</code>	$L > 1$, or $0 < L \leq 1$ and $m^2 d_a^2 L > (1 - L)^2$
<code>product_logcap</code>	$md_a > M_+$ when M_+ is available
<code>critical_variance_negative</code>	$V_0 < 0$ or $V < 0$
<code>moment_A_packed</code>	$d_a^2 > (a_+^2 + V - 1)_+(1 - s_-)$
<code>moment_B_packed</code>	$d_h^2 > V v_+$
<code>near_uniform</code>	$r_- \geq 99999/100000$

$0 \leq H \leq 1$ and $b(t) \geq 0$ on $[0, 1]$. When $A > 0$, the possible interior minimum occurs at $t = (H + A)/(2A)$ and equals $1 - (H + A)^2/(4A)$, so this check is rational and exact.

Define

$$\kappa^2 = \min\{1 - H, V s_+\}, \quad d^2 = \max\{0, \min\{V v_+, 1 - H - a_-^2 v_-\}\}, \quad \ell(t) = 1 - t + \kappa t. \quad (\text{A.3})$$

Then (4.2) holds and $d \geq |1 - h|$. The latter follows from $|1 - h|^2 \leq V v$ and $|1 - h|^2 = \beta(1) - a^2 v$. On a box containing admissible data, both quantities inside the inner minimum are nonnegative; the positive part therefore does not weaken the assertion.

Let Φ_+ be the maximum of Φ_m over $[a_-, a_+] \cap [0, \sqrt{m}]$. A box with empty intersection has already been excluded. Monotonicity in Lemma 3.3 shows that the maximum is attained at 1 if $1 \in [a_-, a_+]$, at a_+ if $a_+ < 1$, and at a_- if $a_- > 1$. Thus its evaluation requires no irrational interval endpoint. The interval upper bound

$$\mathcal{C}_+ = \frac{\Phi_+ r_+ s_+^{m/2}}{T_{32}(\lambda_-/2)}$$

follows from (3.8) and (A.1). A box entirely outside $a^2 \leq m$ is already excluded, so every use of Φ_+ has a nonnegative base.

The tests `direct` and `direct_retained` evaluate (4.7), using (4.3) alone or also (4.4). The tests `center_absolute` and `center_ratio` evaluate (4.8) and (4.9); they require $m \geq 4$. Integrals are bounded by (4.10) and (4.11), with the constants in (4.1). On $I = [l, u]$, a valid constant lower bound for $|D(t)|$ is

$$\max\{0, 1 - a_+ r_+ u, a_- r_- l - 1\}. \quad (\text{A.4})$$

The nonsingular estimates are always retained. Every coefficient outside an integral is replaced by an upper bound, for example $K \leq (m + 1)a_+^3 r_+$ times the computed bound for $\int t^2 \mathcal{U}$.

The test `center_coeff` uses (7.3) with $\mathcal{C}_+$, d from (A.3), and upper endpoints in every nonnegative coefficient. The final test `center_positive` uses Lemma 7.5, in the form (7.10). It requires $r_- > 0$ and $h_+ := a_+ r_+ \leq 1$. Put $h_- = a_- r_-$ and

$$\begin{aligned} t_0 &= \max\{0, \eta, 1 - m(1 - s_-)\}, \\ R_- &= \max\left\{-1, m r_- - (m - 1), \frac{m^2 r_-^2 + t_0 - (m - 1)^2}{2 m r_-}\right\}, \\ W_+ &= \min\{(m - 1)(1 - r_-)^2, (1 - r_-)(r_+ - R_-)\}, \quad L_2 = \frac{m}{2}(v_- - 2W_+). \end{aligned}$$

Here $\eta = 0$ when the refinement in (A.2) is unavailable. Each entry in (7.6) is nondecreasing in $r > 0$ for $0 \leq t_0 \leq 1$ and $m \geq 3$. Lemma 7.4 therefore gives $W \leq W_+$, and the checker requires $L_2 > 0$. Finally set

$$x_- = \frac{H + a_-^2 s_-}{2a_+}, \quad \varepsilon = \sqrt{\max\{0, 2(1 - x_-/r_+)\}}, \quad K_- = (m+1)h_-I_{2,m-2}(1 - h_+).$$

The identity $2ax = \rho + a^2s$ gives $x \geq x_-$. Hence $\varepsilon \geq |e^{i\theta} - 1|$. Also $0 \leq K_- \leq K_2(ar)$. In (7.9), replace h_0 by h_+ to bound E_2 from above. In (7.10), the negative contribution is bounded from above by $-a_-^2 K_- L_2$. It is rounded in the direction appropriate for a subtraction, as specified in Appendix B.

A.2. Several consecutive degrees in one block. For $m \in [M, N]$, where $13 \leq M \leq N$, the second checker uses only Lemma 3.1, Lemma 3.3, and the integral bounds. Set

$$V = 1 - \frac{1}{N}, \quad v_+ = s_+ - r_-^2, \quad H = \max\{r_-^2 + (1 - s_+)/N, 1 - (a_+^2 + V)v_+\}, \quad A = a_-^2 s_-.$$

Use $b(t) = 1 - Ht - At(1 - t)$ and $\kappa^2 = \min\{1 - H, Vs_+\}$, with the same nonnegativity checks as above. The elementary tests are the degree bound $a_-^2 > N$, the nonnegative variance test, the two covariance tests with this V , the inverse-moment test, the negative- b test, and the product test described below.

Lemma 3.3 gives the uniform bound

$$|hJ| \leq \mathcal{C}_+ := \max_{\substack{a \in [a_-, a_+] \\ a^2 \leq N}} \Phi_N(a) r_+ s_+^{M/2}.$$

The maximum is evaluated by the same endpoint rule as in Appendix A.1, with N in place of m . For the elementary product test, let $B = \frac{M+1}{M}(1 + a_+^2)s_+$. A sufficient contradiction is $B^N < (M+1)^2$. To justify the exponent when $B < 1$, note that any admissible point would already require its own product base in (3.9) to exceed one. Thus $B \leq 1$ itself excludes admissible data. When $B > 1$, increasing the exponent to N is an upper bound.

In the direct integral bound, use coefficients $\min\{1, (5/3)s_+\}$ and $\min\{1, 3s_+\}$, powers $b^{(M-1)/2}$ and ℓ^{M-1} , and the prefactor Na_+^2 . The endpoint terms are κ^M and $[N/(N+1)]\mathcal{C}_+$. These are upper bounds because $0 \leq b, \ell, \kappa \leq 1$.

For `direct_retained`, fix δ_I as in Lemma 4.1. On admissible data, $b(t) \geq \delta_I^2$ and $\ell(t) \geq \delta_I$. Therefore $(mb(t) - \delta_I^2)/(m-1)$ and $(m\ell(t) - \delta_I)/(m-1)$ are nonincreasing in m . Bound their bases at $m = M$. These enlarged bases may exceed one. For that reason, at each endpoint the checker uses

$$\max\{B^{(M-1)/2}, B^{(N-1)/2}\}, \quad \max\{L^{M-1}, L^{N-1}\}, \quad (\text{A.5})$$

where B, L denote the respective nonnegative enlarged bases at that endpoint. The corresponding functions of t are convex, being maxima of two convex powers. Applying (4.10) to them is therefore valid. An interval contains admissible data only where the preceding monotonicity conditions hold; using positive parts outside that set does not decrease these majorants on admissible data.

For centered integrals, use the four upper coefficients $5N/(6d_I)$, $3N/(2d_I)$, $3N(N-1)/2$, and $9N(N-1)/2$, where d_I is the bound in (A.4). The first two are omitted if $d_I = 0$. Their powers are respectively $b^{(M-1)/2}$, ℓ^{M-1} , $b^{(M-2)/2}$, and ℓ^{M-2} . The prefactor in K is $(N+1)a_+^3 r_+$. For (4.8), use the endpoint bound $(Vv_+)^{(M+1)/2}$. For (4.9), use $(1 + r_+)\{V^{(M+1)/2}(1 - r_-^2)^{(M-1)/2} + K_+\}$. All bases in these centered powers belong to $[0, 1]$; decreasing the exponent therefore enlarges them. Formula (A.5) is the deliberate exception for bases not known to be at most one.

APPENDIX B. EXACT ARITHMETIC AND REPRODUCIBILITY

B.1. Certificates and coverage. The two JSON files encode the partitions used in Proposition 8.2. A degree block records the inclusive range $[M, N]$ for $m = n - 1$, a positive mesh refinement, and a list of leaf paths with test identifiers. The initial parameter box is (8.1). Each pair of characters in a path consists of a coordinate index in $\{0, 1, 2\}$ and a child index in $\{0, 1\}$; the corresponding coordinate interval is bisected at its midpoint. The lower and upper children are both closed.

The checker inserts the paths into a tree. It rejects repeated leaves, a leaf with descendants, conflicting split coordinates, and any internal node without both children. It reconstructs all endpoints as exact rational numbers. This proves coverage by induction on the finite tree and ensures that leaf interiors are disjoint. Consecutive degree blocks are checked for both omissions and overlaps. Thus no information produced by the floating-point search is trusted merely because it appears in a certificate.

For a block beginning at $m = M$, set $L = \max\{4, \lceil \log_2 M \rceil + 2\}$. The mesh first divides $[0, 2^{-L}]$ into q equal parts, and then divides each interval $[2^{-j}, 2^{-j+1}]$, $j = L, \dots, 1$, into q equal parts. The small-degree certificates use $q = 32$. The degree-14 certificate uses $q = 16$, and the other blocks use $q = 4$. The checker verifies directly that the resulting rational nodes increase from zero to one. Convex quadrature is precisely (4.10)–(4.11), not a floating-point quadrature rule.

B.2. Arithmetic operations. Fix $S = 2^b$, where the primary run uses $b = 100$. A nonnegative integer U represents an upper bound U/S . For a nonnegative rational p/q , its enclosure is $\lceil Sp/q \rceil$. Multiplication of upper bounds is implemented as

$$(U, V) \mapsto \left\lceil \frac{UV}{S} \right\rceil, \quad \left(\frac{p}{q}, U \right) \mapsto \left\lceil \frac{pU}{q} \right\rceil. \quad (\text{B.1})$$

For a square root, choose the smallest integer K such that $K^2 \geq US$; then $K/S \geq \sqrt{U/S}$. It is obtained from the integer square root of US , with one added unless that product is a perfect square. Integer powers use repeated squaring with (B.1); half-integer powers use an integer power and one square root. All operands and denominators are checked for their required signs.

The quantities subtracted in (7.10) are treated differently. If $c \geq 0$ is a rational lower bound for the positive contribution $a^2 K_{-L_2}$, the checker subtracts $\lfloor Sc \rfloor$, not $\lceil Sc \rceil$, from the upper-bound numerator. Thus directed rounding cannot make the final upper bound artificially small. The exponential factors are bounded through the exact positive Taylor polynomial T_{32} defined after Table 2; no floating-point exponential or logarithm is used for acceptance.

The invariance of the upper-bound interpretation under these operations follows immediately by induction. A main test is accepted only if its final integer numerator U satisfies $U < S$. The positive integer margins in (8.2) are therefore strict inequalities, not tolerance estimates. Elementary exclusions are comparisons of exact rational numbers. Timing measurements in the logs are irrelevant to acceptance.

B.3. Reproduction. The essential supplementary files are the two certificate files and `verify_low.py`, `verify_mid.py`, and `verify_all.py`. The first checker treats $4 \leq n \leq 13$ using Appendix A.1; the second treats $14 \leq n \leq 100000$ using Appendix A.2. The first imports the integer-arithmetic routines and mesh construction of the second, but neither imports a generator. The wrapper invokes both, checks their endpoints, and records the combined result. In a directory containing these files, run

```
python3 self_test.py
python3 verify_all.py --bits 100
python3 verify_all.py --bits 160
```

Only the Python standard library is needed for these commands. The optional generators require NumPy, but are not part of the acceptance procedure.

The certificate files are identified by the following SHA-256 digests; line breaks are not part of the digests.

```
certificate_n4_to_n13.json
4334dd70fe5317040124115239f92ecc
4e220cbf6c3576407396c3ea9df19f8b

certificate_n14_to_n100000.json
f9131604a764777836afed040bd73b1f
8ca24c5696022460b5547cb509af766d
```

For the first certificate the numbers of leaves, in degrees 4 through 13, are respectively 532, 1103, 1594, 2540, 1753, 1420, 1164, 1002, 904, 809. Ten leaves, five in degree four and five in degree five, invoke Lemma 7.6. All other leaves are checked by the rational or integer tests in Appendix A.

The complete runs at 100 and 160 bits verify the same partitions. They are checks at different precisions, not two independent implementations of the entire proof. Neither run is a proof-assistant formalization of the mathematical lemmas. The supplementary arithmetic self-tests verify elementary identities, selected analytic constants, and rejection of malformed or false certificates; sample tests are not used as replacements for any mathematical argument.

B.4. Elementary analytic constants. For completeness, the exponential comparisons in Proposition 5.1 can be checked entirely with rational numbers. If $x \geq 0$ and $x < k + 2$, then

$$\sum_{j=0}^k \frac{x^j}{j!} \leq e^x \leq \sum_{j=0}^k \frac{x^j}{j!} + \frac{x^{k+1}}{(k+1)!} \frac{1}{1 - x/(k+2)}. \quad (\text{B.2})$$

The upper bound follows because the ratios of successive tail terms do not exceed $x/(k+2)$. Formula (B.2) with $k = 20$ verifies all four comparisons in Step 2 of that proposition. It also gives $e < 11/4 < 25/9$, which proves the uniform bounds following (4.1). The remaining estimates in Proposition 5.1 follow from the displayed rational inequalities, $1 - u \leq e^{-u}$ for $u \geq 0$, and $e^x \geq x^6/720$ for $x \geq 0$. These are analytic estimates valid throughout the stated range of m , not extrapolations of finite checks.

ACKNOWLEDGMENTS

The author is grateful to Terence Tao for helpful discussions in the comments on his blog post [Tao26]. The author is supported by the China Scholarship Council, the Young Elite Scientists Sponsorship Program for PhD Students of the China Association for Science and Technology, and the Fundamental Research Funds for the Central Universities at Xi'an Jiaotong University (Grant No. xzy022024045).

Use of AI tools. ChatGPT was used for language editing, bibliographic checks, assistance with reviewing and reorganizing the arguments, and preparation of the explanatory diagram. The finite certificates are checked by the exact-arithmetic programs specified in Appendix B.

DATA AVAILABILITY

The certificates, exact checkers, optional generators, and verification logs are available in the supplementary repository [Zha26S]. This paper uses the files at [commit 83ee46a7c971](#); the certificate digests are recorded in Appendix B. These certificates are an essential part of Proposition 8.2. Reproduction requires only the Python standard library.

REFERENCES

- [BD00] R. Bhatia and C. Davis, *A better bound on the variance*, Amer. Math. Monthly **107** (2000), no. 4, 353–357. [10.1080/00029890.2000.12005203](https://doi.org/10.1080/00029890.2000.12005203).
- [Bor96] I. Borcea, *The Sendov conjecture for polynomials with at most seven distinct zeros*, Analysis **16** (1996), no. 2, 137–160. [10.1524/analy.1996.16.2.137](https://doi.org/10.1524/analy.1996.16.2.137).
- [Bor07] J. Borcea, *Equilibrium points of logarithmic potentials induced by positive charge distributions. I. Generalized de Bruijn–Springer relations*, Trans. Amer. Math. Soc. **359** (2007), no. 7, 3209–3237. [10.1090/S0002-9947-07-04251-1](https://doi.org/10.1090/S0002-9947-07-04251-1).
- [BX99] J. E. Brown and G. Xiang, *Proof of the Sendov conjecture for polynomials of degree at most eight*, J. Math. Anal. Appl. **232** (1999), no. 2, 272–292. [10.1006/jmaa.1999.6267](https://doi.org/10.1006/jmaa.1999.6267).
- [Deg14] J. Dégot, *Sendov conjecture for high degree polynomials*, Proc. Amer. Math. Soc. **142** (2014), no. 4, 1337–1349. [10.1090/S0002-9939-2014-11888-0](https://doi.org/10.1090/S0002-9939-2014-11888-0).
- [Hay67] W. K. Hayman, *Research problems in function theory*, The Athlone Press, University of London, London, 1967.
- [KPPSS11] D. Khavinson, R. Pereira, M. Putinar, E. B. Saff, and S. Shimorin, *Borcea’s variance conjectures on the critical points of polynomials*, in *Notions of positivity and the geometry of polynomials* (P. Brändén, M. Passare, and M. Putinar, eds.), Trends Math., Birkhäuser, Basel, 2011, pp. 283–309. [10.1007/978-3-0348-0142-3_16](https://doi.org/10.1007/978-3-0348-0142-3_16).
- [Mal05] S. M. Malamud, *Inverse spectral problem for normal matrices and the Gauss–Lucas theorem*, Trans. Amer. Math. Soc. **357** (2005), no. 10, 4043–4064. [10.1090/S0002-9947-04-03649-9](https://doi.org/10.1090/S0002-9947-04-03649-9).
- [Mar66] M. Marden, *Geometry of polynomials*, 2nd ed., Math. Surveys, vol. 3, American Mathematical Society, Providence, RI, 1966.
- [Maz26] L. Mazur, *A computer-assisted proof of Sendov’s conjecture*, preprint, August 5, 2026, 19 pp. https://www.proofatlas.ai/papers/sendov-conjecture/SENDOV_CONJECTURE_PROOF_AUGUST_5_2026.pdf.
- [MS69] A. Meir and A. Sharma, *On Ilyeff’s conjecture*, Pacific J. Math. **31** (1969), no. 2, 459–467. [10.2140/pjm.1969.31.459](https://doi.org/10.2140/pjm.1969.31.459).
- [Per03] R. Pereira, *Differentiators and the geometry of polynomials*, J. Math. Anal. Appl. **285** (2003), no. 1, 336–348. [10.1016/S0022-247X\(03\)00465-7](https://doi.org/10.1016/S0022-247X(03)00465-7).
- [RS02] Q. I. Rahman and G. Schmeisser, *Analytic theory of polynomials*, London Math. Soc. Monogr. (N.S.), vol. 26, The Clarendon Press, Oxford University Press, Oxford, 2002.
- [Sch86] I. J. Schoenberg, *A conjectured analogue of Rolle’s theorem for polynomials with real or complex coefficients*, Amer. Math. Monthly **93** (1986), no. 1, 8–13. [10.1080/00029890.1986.11971734](https://doi.org/10.1080/00029890.1986.11971734).
- [TZ25] Q. Tang and T. Zhang, *Sharp Schoenberg type inequalities and the de Bruijn–Sharma problem*, preprint, 2025, [arXiv:2508.10341v3](https://arxiv.org/abs/2508.10341v3).
- [Tao22] T. Tao, *Sendov’s conjecture for sufficiently-high-degree polynomials*, Acta Math. **229** (2022), no. 2, 347–392. [10.4310/ACTA.2022.v229.n2.a3](https://doi.org/10.4310/ACTA.2022.v229.n2.a3).
- [Tao26] T. Tao, *A digestion of the proof of Sendov’s conjecture*, What’s New, August 12, 2026. <https://terrytao.wordpress.com/2026/08/12/a-digestion-of-the-proof-of-sendovs-conjecture/>.
- [Zha26] T. Zhang, *Beyond Sendov’s conjecture: the quadratic Tang–Zhang inequality*, preprint, 2026, [arXiv:2609.19126v1](https://arxiv.org/abs/2609.19126v1).
- [Zha26S] T. Zhang, *Supplementary code and certificates for Borcea’s 2-variance conjecture*, version September 29, 2026. <https://github.com/zhangteng2000/borcea-variance>.

SCHOOL OF MATHEMATICS AND STATISTICS, XI’AN JIAOTONG UNIVERSITY, XI’AN, CHINA